Introduction

Using an open dataset (csv format) from **"Kaggle website"**, to understand the main factors that cause diabetes. And analyze data to find out what might lead to diabetes by comparing different age groups, blood pressure, and glucose levels. This analysis will help us conclude which groups are more likely to get diabetes now or could be at risk in the future.

Objective

1. To explore and analyze the relationships between various health parameters (such as glucose levels, BMI, age, insulin levels, and blood pressure) and the occurrence of diabetes.
2. To visualize data trends and identify correlations to enhance understanding of factors contributing to diabetes.
3. To develop insights that could help in predicting or preventing diabetes based on health indicators.

```python
data = pd.read_csv("diabetes.csv")
data.info()
```

[12] ✓ 0.0s

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Glucose        768 non-null    int64
 1   BloodPressure  768 non-null    int64
 2   Insulin        768 non-null    int64
 3   BMI            768 non-null    float64
 4   Age            768 non-null    int64
 5   Outcome        768 non-null    int64
dtypes: float64(1), int64(5)
memory usage: 36.1 KB
```

The original dataset includes columns like pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and outcome. But to keep things simple, we will only focus on a few columns mentioned below. As we will create some graphs to understand different concepts and see how these factors influence diabetes.

Original Dataset Src : [Diabetes Dataset For Beginners]

Data columns

1. **Glucose:** Glucose level in the blood (int64)
2. **BloodPressure:** Blood pressure measurement (int64)
3. **Insulin:** Insulin level in the blood (int64)
4. **BMI:** Body Mass Index (BMI) (float64)
5. **Age:** Age of the person (int64)
6. **Outcome:** Indicates if the person has diabetes (1 for diabetic, 0 for non-diabetic) (int64)

```
data.head()
```

[53] ✓ 0.0s

	Glucose	BloodPressure	Insulin	BMI	Age	Outcome
0	148	72	0	33.6	50	1
1	85	66	0	26.6	31	0
2	183	64	0	23.3	32	1
3	89	66	94	28.1	21	0
4	137	40	168	43.1	33	1

Reasoning

1. Diabetes is a major health concern globally, and understanding how different health factors correlate with its occurrence is crucial for early detection and prevention.
2. The analysis aims to provide a better understanding of how parameters like glucose, BMI, and insulin levels are related to the likelihood of being diabetic.
3. By visualizing these relationships, the project could help in forming strategies for better healthcare management and awareness.

Problem Statement

1. Given a dataset with various health indicators and diabetes outcomes, the task is to analyze and visualize the correlations between these indicators and the likelihood of being diabetic.
2. The goal is to explore patterns and uncover relationships that might not be immediately apparent through raw data alone.

Modules and Libraries Used

1. **Pandas**: For data manipulation and analysis (e.g., cleaning, categorization, calculating correlations).
2. **Matplotlib**: For data visualization, including bar graphs, histograms, and pie charts.
3. **Seaborn**: For advanced visualizations such as pair plots and correlation heatmaps.

Requirements

1. **Hardware Requirements**
    a. Any system capable of running Python, with at least 4GB of RAM.
2. **Software Requirements**:
    a. Python (preferably 3.x version).
    b. Libraries: Pandas, Matplotlib, Seaborn (can be installed via pip install).
    c. A Python IDE (e.g., Jupyter Notebook, VSCode, or PyCharm).

```python
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv("diabetes.csv")
```

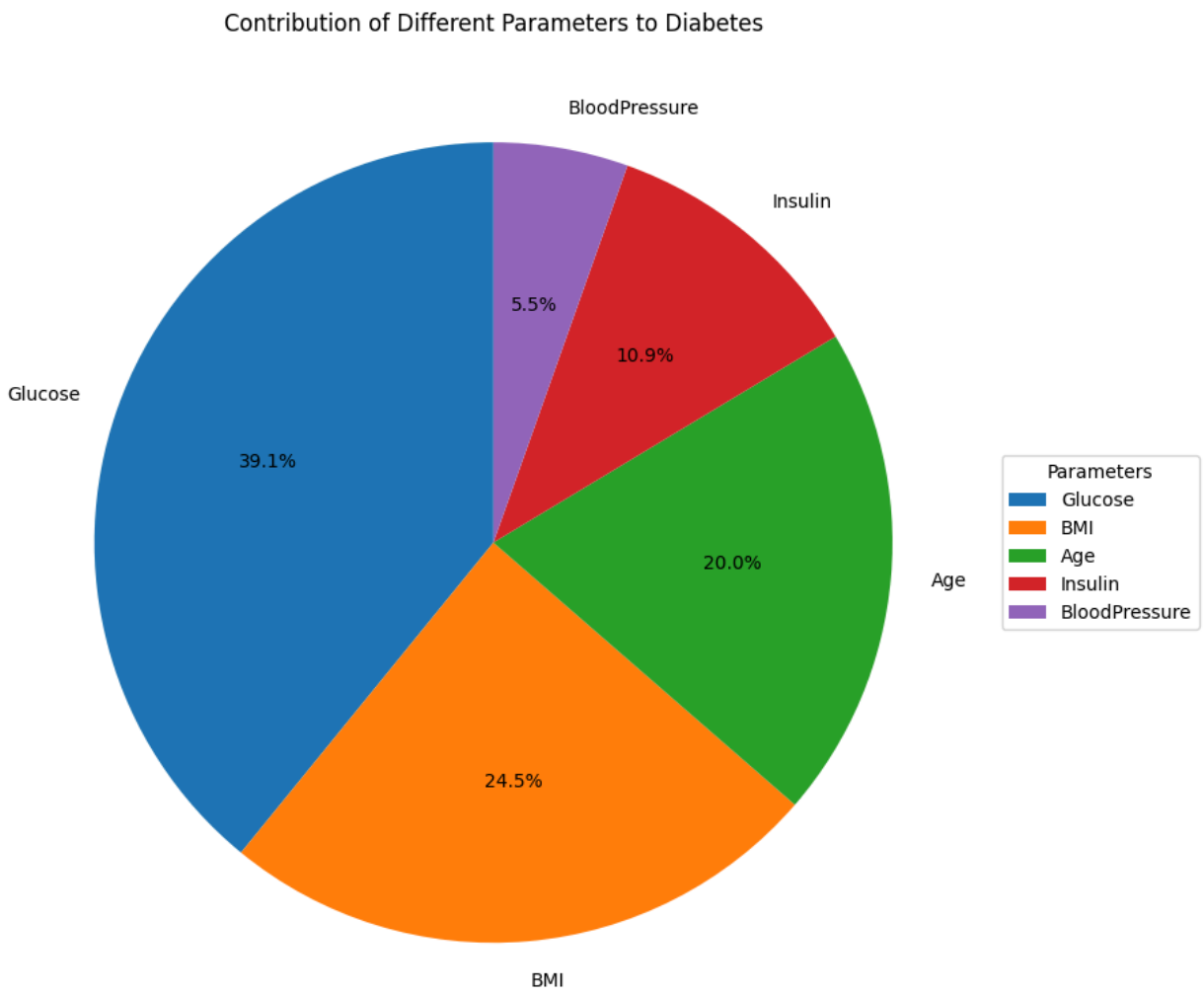
To start with the analyzing part we will use pandas, matplotlib and seaborn initially we will import the modules and will fetch the data from the csv file by the help of pd (pandas) module.

Methodology

- **Data Preprocessing**: Importing, cleaning, and categorizing data (e.g., categorizing glucose, BMI, blood pressure levels into ranges).
- **Exploratory Data Analysis (EDA)**: Using statistical methods (correlation) and visualizations (scatter plots, pair plots, bar charts) to uncover insights.
- **Statistical Analysis**: Calculation of correlations between health parameters and diabetes outcome, followed by sorting and visualizing the most influential factors.
- **Visualization**: Generating various graphs like bar charts, histograms, and pair plots to present findings.

1. **Generate a pie chart to understand the correlation of factors with diabetes and which contributes the most to diabetes.**

```python
# Calculate absolute correlations with Outcome
parameters = ['Glucose', 'BMI', 'Age', 'BloodPressure', 'Insulin']
correlations = data[parameters + ['Outcome']].corr()['Outcome'].abs()
correlations = correlations[:-1]  # Remove Outcome's self-correlation

# Sort correlations
correlations_sorted = correlations.sort_values(ascending=False)

# Create pie chart
plt.figure(figsize=(10, 8))
plt.pie(correlations_sorted, labels=correlations_sorted.index,
        autopct='%1.1f%%', startangle=90)

# Add title
plt.title('Contribution of Different Parameters to Diabetes')

# Add legend
plt.legend(correlations_sorted.index, title="Parameters",
           loc="center left", bbox_to_anchor=(1, 0, 0.5, 1))

plt.tight_layout()
plt.show()
```



Correlation values with Diabetes
Glucose: 0.467
BMI: 0.293
Age: 0.238
Insulin: 0.131
BloodPressure: 0.065

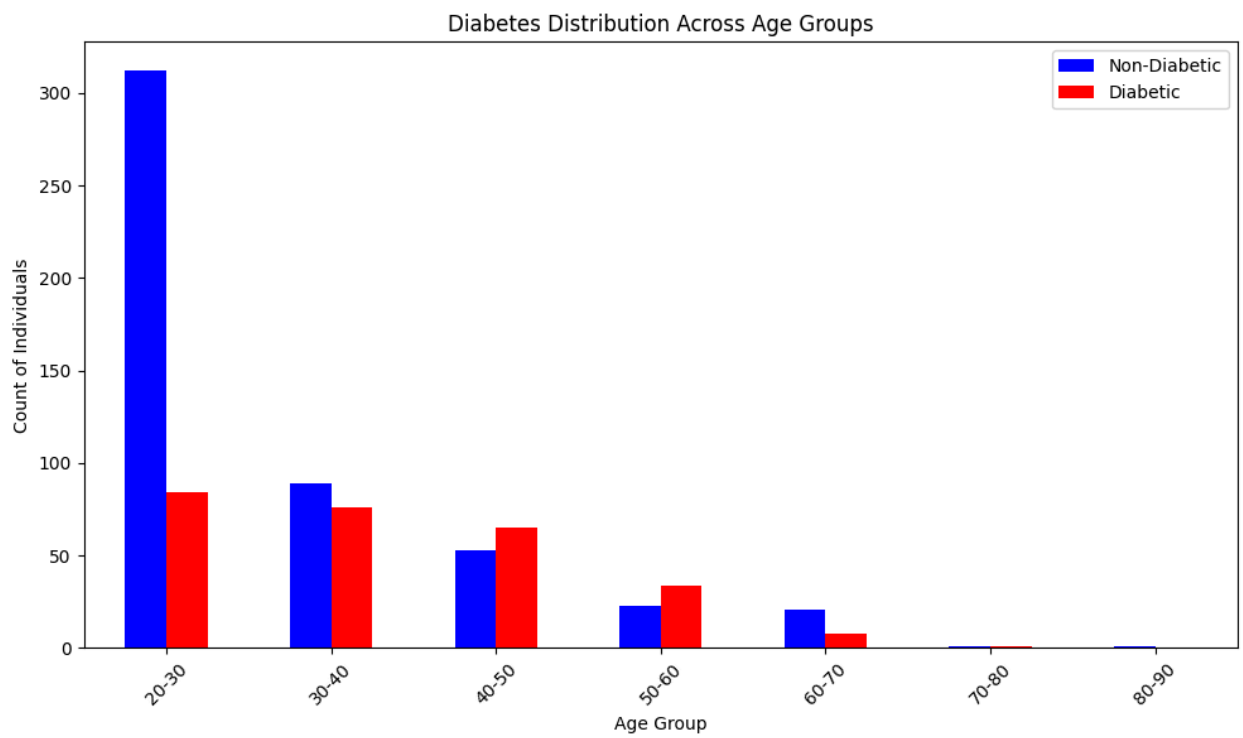
If we follow the correlation values and pie chart above we can clearly see that the contribution of Glucose, BMI, and Age is the most when it comes to diabetes.

Now why?

- Glucose: Most direct indicator of diabetes.
- BMI: Strong risk factor for Type 2 diabetes.
- Age: Significant risk factor for diabetes development.

2. Generate a bar graph to see the Diabetes distribution across age groups.

```python
# Define age groups for categorization
age_bins = [20, 30, 40, 50, 60, 70, 80, 90]  # Example age groups
age_labels = ['20-30', '30-40', '40-50', '50-60', '60-70', '70-80', '80-90']
# Labels for each group

# Add a new column 'AgeGroup' to categorize individuals into age groups
data['AgeGroup'] = pd.cut(data['Age'], bins=age_bins, labels=age_labels, right=False)

# Count how many diabetic (Outcome=1) and non-diabetic (Outcome=0) individuals are in each age group
age_group_counts = data.groupby(['AgeGroup', 'Outcome']).size().unstack(fill_value=0)

# Plot the bar graph
age_group_counts.plot(kind='bar', stacked=False, color=['blue', 'red'], figsize=(10, 6))

# Set plot labels and title
plt.xlabel('Age Group')
plt.ylabel('Count of Individuals')
plt.title('Diabetes Distribution Across Age Groups')
plt.xticks(rotation=45)  # Rotate x-axis labels for better readability
plt.legend(['Non-Diabetic', 'Diabetic'])
plt.tight_layout()

# Show the plot
plt.show()
```

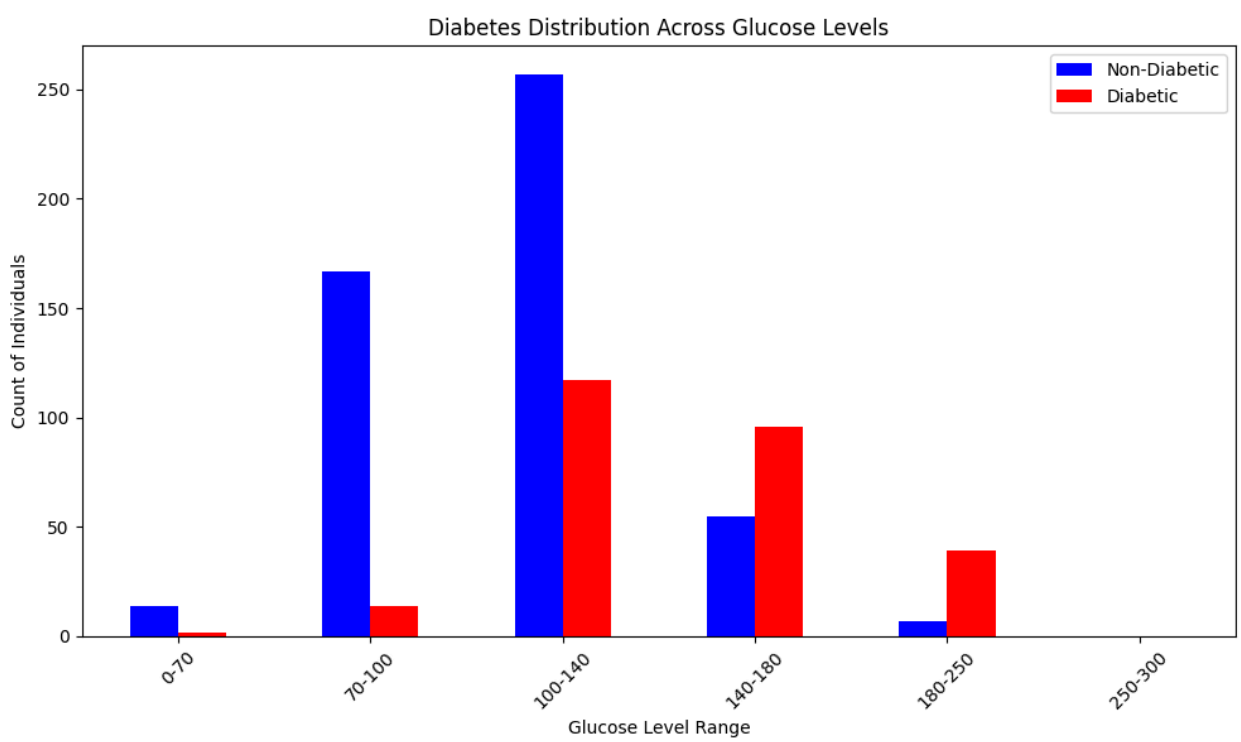


With reference to the graph above we can say :

1. Most of the People between the age group 20-30 are Non-diabetic.
2. Between the age group 20 to 50 are more likely to get diabetes.
3. People between the age group 50-60 more people are diabetic as compared to people who are Non-diabetic.

3. Generate a bar graph to see the Diabetes distribution across Glucose Levels.

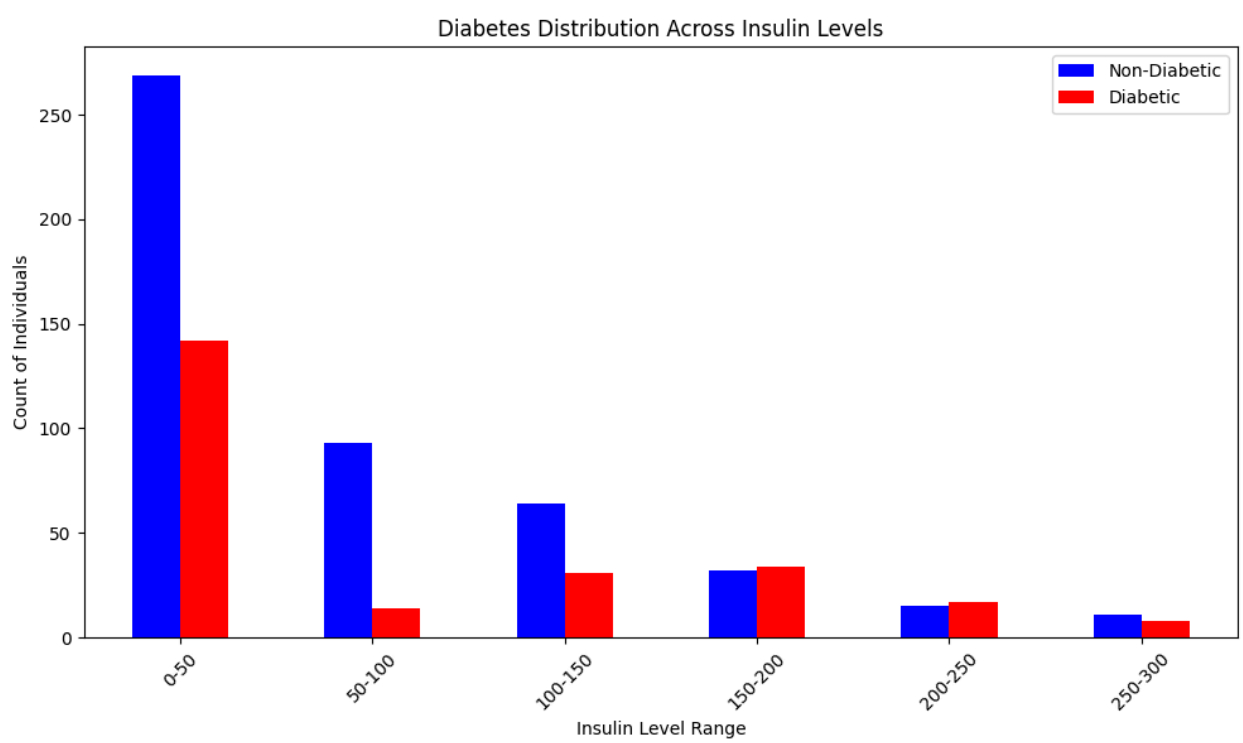
```python
# Define glucose level ranges for categorization
glucose_bins = [0, 70, 100, 140, 180, 250, 300]  # Example glucose ranges
glucose_labels = ['0-70', '70-100', '100-140', '140-180', '180-250', '250-300']  # Labels for each range

# Add a new column 'GlucoseGroup' to categorize individuals into glucose groups
data['GlucoseGroup'] = pd.cut(data['Glucose'], bins=glucose_bins, labels=glucose_labels, right=False)

# Count how many diabetic (Outcome=1) and non-diabetic (Outcome=0) individuals are in each glucose group
glucose_group_counts = data.groupby(['GlucoseGroup', 'Outcome']).size().unstack(fill_value=0)

# Plot the bar graph
glucose_group_counts.plot(kind='bar', stacked=False, color=['blue', 'red'], figsize=(10, 6))

# Set plot labels and title
plt.xlabel('Glucose Level Range')
plt.ylabel('Count of Individuals')
plt.title('Diabetes Distribution Across Glucose Levels')
plt.xticks(rotation=45)  # Rotate x-axis labels for better readability
plt.legend(['Non-Diabetic', 'Diabetic'])
plt.tight_layout()

# Show the plot
plt.show()
```



With reference to the graph above we can say that people who have a glucose level between 100 to 250 are more prone to get diabetic. And people with less than 100 are less prone to diabetic.

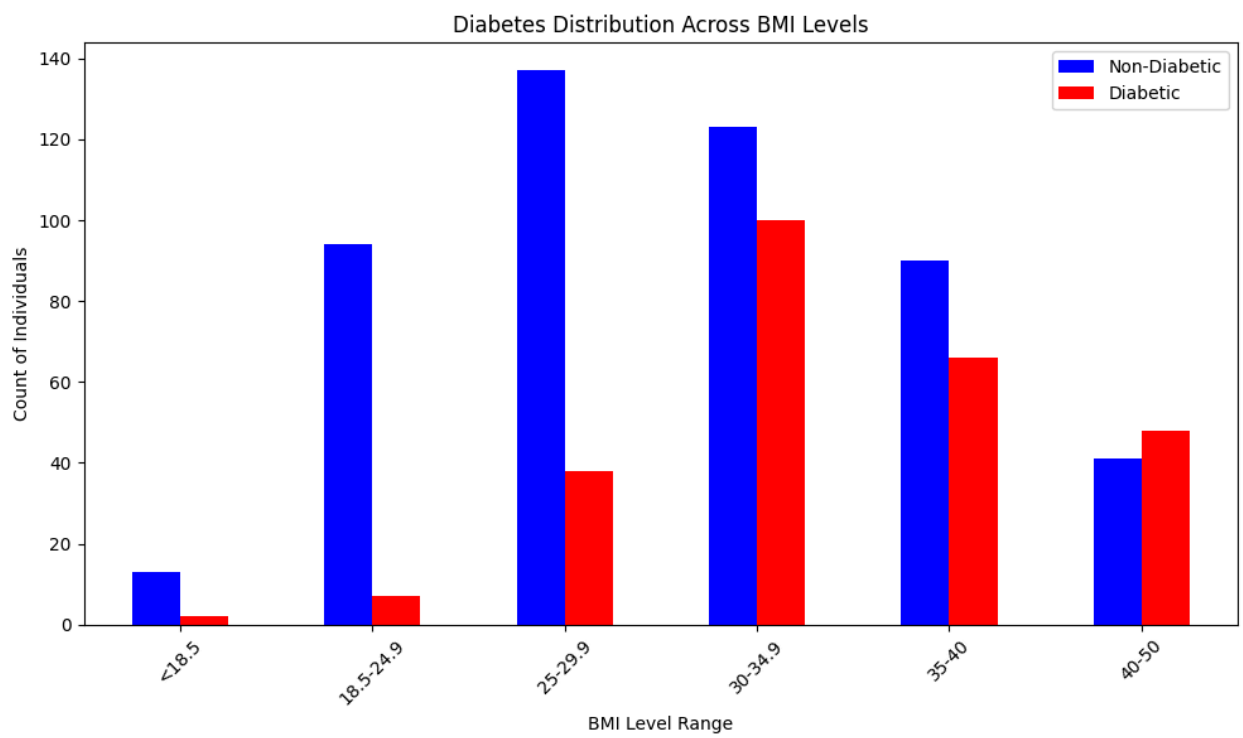
4. **Generate a bar graph to see the Diabetes distribution across Insulin levels.**

```python
# Define insulin level ranges for categorization
insulin_bins = [0, 50, 100, 150, 200, 250, 300]  # Example insulin ranges
insulin_labels = ['0-50', '50-100', '100-150', '150-200', '200-250', '250-300']  # Labels for each range

# Add a new column 'InsulinGroup' to categorize individuals into insulin groups
data['InsulinGroup'] = pd.cut(data['Insulin'], bins=insulin_bins, labels=insulin_labels, right=False)

# Count how many diabetic (Outcome=1) and non-diabetic (Outcome=0) individuals are in each insulin group
insulin_group_counts = data.groupby(['InsulinGroup', 'Outcome']).size().unstack(fill_value=0)

# Plot the bar graph
insulin_group_counts.plot(kind='bar', stacked=False, color=['blue', 'red'], figsize=(10, 6))

# Set plot labels and title
plt.xlabel('Insulin Level Range')
plt.ylabel('Count of Individuals')
plt.title('Diabetes Distribution Across Insulin Levels')
plt.xticks(rotation=45)  # Rotate x-axis labels for better readability
plt.legend(['Non-Diabetic', 'Diabetic'])
plt.tight_layout()

# Show the plot
plt.show()
```



5. **Generate a bar graph to see the Diabetes distribution across BMI levels.**

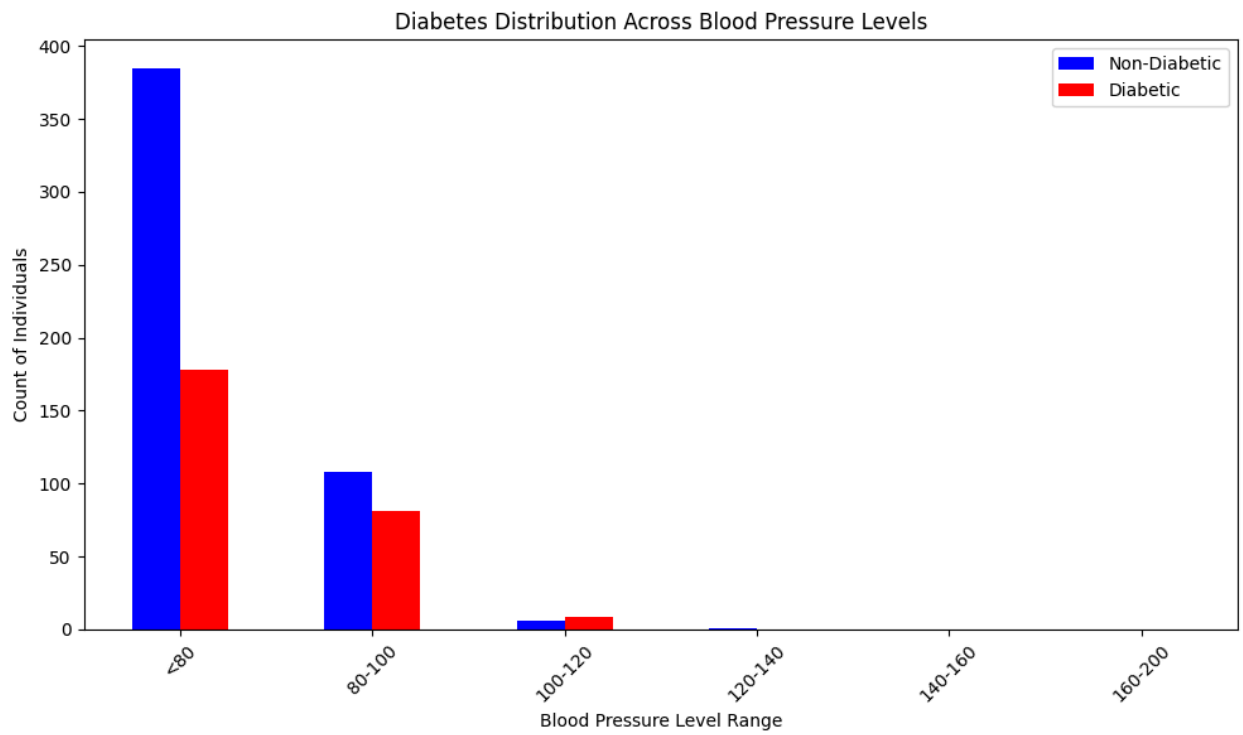
```python
# Define BMI level ranges for categorization
bmi_bins = [0, 18.5, 24.9, 29.9, 34.9, 40, 50]
# Example BMI ranges (underweight, normal, overweight, obese, etc.)
bmi_labels = ['<18.5', '18.5-24.9', '25-29.9', '30-34.9', '35-40', '40-50']  # Labels for each range

# Add a new column 'BMIGroup' to categorize individuals into BMI groups
data['BMIGroup'] = pd.cut(data['BMI'], bins=bmi_bins, labels=bmi_labels, right=False)

# Count how many diabetic (Outcome=1) and non-diabetic (Outcome=0) individuals are in each BMI group
bmi_group_counts = data.groupby(['BMIGroup', 'Outcome']).size().unstack(fill_value=0)

# Plot the bar graph
bmi_group_counts.plot(kind='bar', stacked=False, color=['blue', 'red'], figsize=(10, 6))

# Set plot labels and title
plt.xlabel('BMI Level Range')
plt.ylabel('Count of Individuals')
plt.title('Diabetes Distribution Across BMI Levels')
plt.xticks(rotation=45)  # Rotate x-axis labels for better readability
plt.legend(['Non-Diabetic', 'Diabetic'])
plt.tight_layout()

# Show the plot
plt.show()
```



With reference to the graph above we can say that people with the BMI range from 30 to 50 are more prone to get diabetic, and people with BMI less than 30 are less prone to get diabetic.

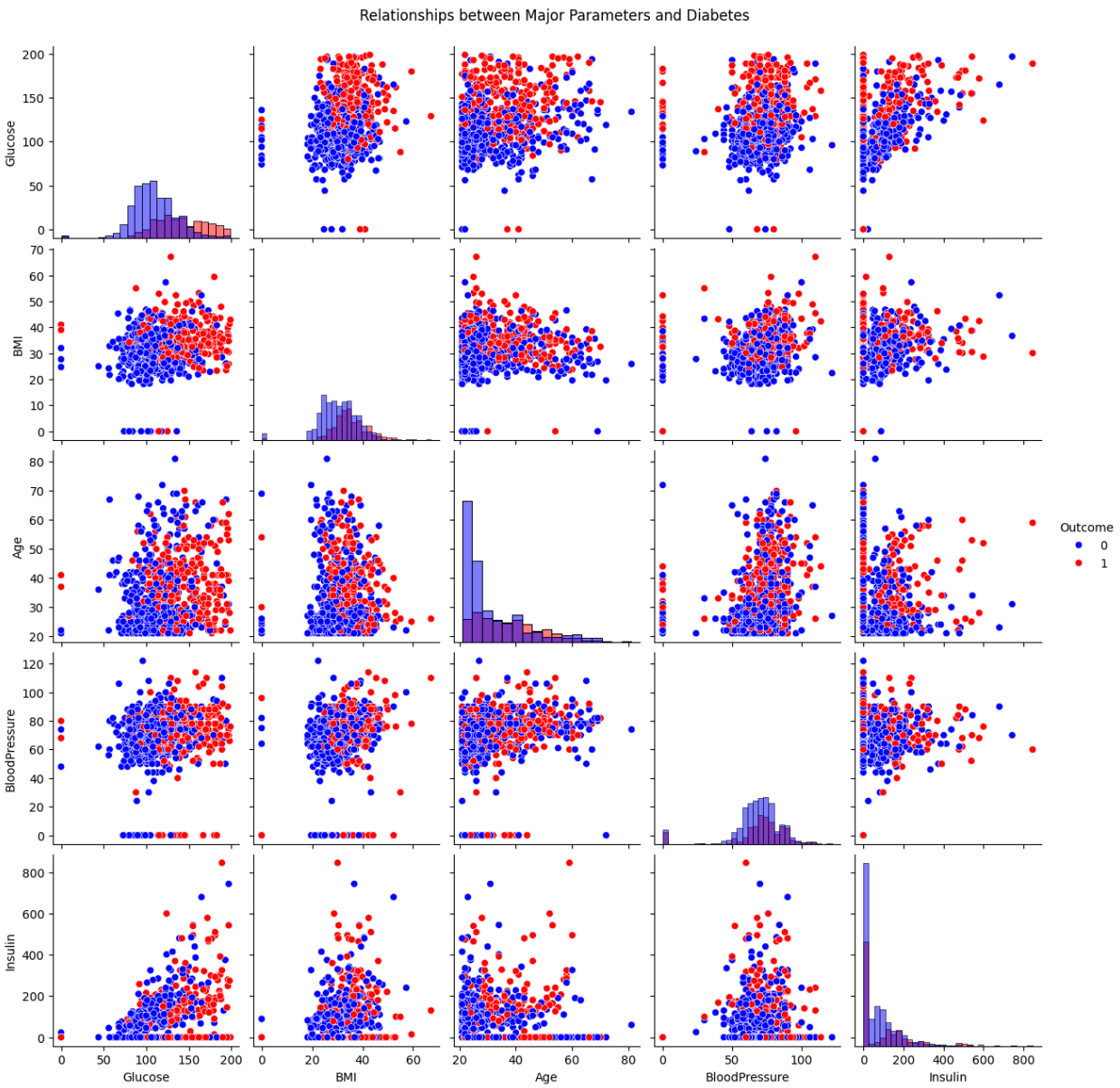
6. **Generate a bar graph to see the Diabetes distribution across BloodPressure levels.**

```python
# Define blood pressure level ranges for categorization
bp_bins = [0, 80, 100, 120, 140, 160, 200]  # Example blood pressure ranges
bp_labels = ['<80', '80-100', '100-120', '120-140', '140-160', '160-200']  # Labels for each range

# Add a new column 'BloodPressureGroup' to categorize individuals into blood pressure groups
data['BloodPressureGroup'] = pd.cut(data['BloodPressure'], bins=bp_bins, labels=bp_labels, right=False)

# Count how many diabetic (Outcome=1) and non-diabetic (Outcome=0) individuals are in each blood pressure group
bp_group_counts = data.groupby(['BloodPressureGroup', 'Outcome']).size().unstack(fill_value=0)

# Plot the bar graph
bp_group_counts.plot(kind='bar', stacked=False, color=['blue', 'red'], figsize=(10, 6))

# Set plot labels and title
plt.xlabel('Blood Pressure Level Range')
plt.ylabel('Count of Individuals')
plt.title('Diabetes Distribution Across Blood Pressure Levels')
plt.xticks(rotation=45)  # Rotate x-axis labels for better readability
plt.legend(['Non-Diabetic', 'Diabetic'])
plt.tight_layout()

# Show the plot
plt.show()
```



7. **Correlations with Diabetes by the help of scatter plot graphs.**

```python
# Select the major parameters we want to analyze
parameters = ['Glucose', 'BMI', 'Age', 'BloodPressure', 'Insulin']

# Create a color palette for diabetic vs non-diabetic
colors = {0 : 'blue', 1 : 'red'}

# Create a pairs plot
plt.figure(figsize=(12, 12))
sns.pairplot(data[parameters + ['Outcome']],
             hue='Outcome',
             palette=colors,
             diag_kind='hist')  # You can also use 'kde' for density plots

plt.suptitle('Relationships between Major Parameters and Diabetes', y=1.02)

# Show the plot
plt.show()
```


Relationships between Major Parameters and Diabetes
Glucose
BMI
Age
BloodPressure
Insulin
Outcome
0
1

Analysis and Outcomes

- Correlation analysis shows that glucose levels have the highest correlation with diabetes outcomes, followed by BMI and age.
- Insights into how higher glucose, BMI, and insulin levels increase the likelihood of diabetes.
- Diabetic individuals tend to fall into higher ranges of glucose, BMI, and insulin.

Conclusion

- The analysis highlights key indicators (glucose, BMI) that significantly affect the likelihood of being diabetic.
- The project emphasizes the need for health monitoring and early intervention, particularly for individuals with higher glucose and BMI values.
- The findings could be used for healthcare professionals to target at-risk individuals and promote preventive measures.

[nayaksomkar/DiabetesDataViz](https://github.com/nayaksomkar/DiabetesDataViz)

[Diabetes Dataset For Beginners]